

SEC P vs NP — notebook summary

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-12 22:00 UTC

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Generated 2026-05-12 22:00 UTC

- Cycles recorded: **677**
- Time span: 456.3h (~1.48 cycles/h)
- Notebook: /home/ludo/Scrivania/SEC/research/pvsnp_notebook.jsonl

Verdict distribution

Verdict	Count
INCONCLUSIVE	640
BARRIER_HIT	18
FALSIFIED	15
SUPPORTED	4

Mathematical fields explored (field_A)

Field	Cycles
Real Algebraic Geometry	37
Schur-Weyl Duality	25
Representation Theory of Symmetric Groups	17
Matroid Theory	16
Free Probability	14
Additive Combinatorics	11
Free Probability Theory	11
Noncommutative L^p Geometry	10
Random Matrix Theory	8
Polymatroid Theory	8
Algebraic Geometry	7
Spectral Graph Theory	7
Finite Geometry	6
Plethysm Theory	6
Noncommutative Harmonic Analysis	5
Ergodic Circuit Framework (communication complexity via dynamical systems)	4
{“framework_name”: “Ergodic Circuit Framework”, “math_branch”: “COMM_COMPLEXITY”}	4
Fourier Analysis on Boolean Functions	4
Fourier Analysis of Boolean Functions	4

Field	Cycles
Algebraic Geometry of Secant Varieties	4
Persistent Homology	4
FOURIER_ANALYSIS	4
Noncommutative Geometry	4
Fourier analysis of boolean functions	3
Algebraic Topology	3
matroid theory	3
Coarse Geometric Lifting (CGL) — Geometric Complexity Theory	3
{“framework_name”: “Tropical Circuit Weight Analysis (TCWA)”, “math_branch”: “BOUNDED_ARITHMETIC”}	3
Tropical Circuit Weight Analysis (BOUNDED_ARITHMETIC)	3
Schur-Weyl duality	3

Last 15 cycles (chronological)

Time	Verdict	Title
2026-05-12 15:52 UTC	INCONCLUSIVE	Additive Energy of Truth Table Bounds ACC^0 Circuit Size for Expli
2026-05-12 16:08 UTC	INCONCLUSIVE	Plethysm Multiplicity Inverse Proportional to Monotone Circuit Si
2026-05-12 16:21 UTC	INCONCLUSIVE	Ehrhart Polynomial Coefficient Sum Bounded by Resolution Proof Si
2026-05-12 17:13 UTC	INCONCLUSIVE	Noncommutative L^∞ Norm Lower Bounds Disjointness Communication C
2026-05-12 17:24 UTC	INCONCLUSIVE	Quasigroup Idempotent Density Inversely Proportional to ACC^0 Circ
2026-05-12 17:37 UTC	INCONCLUSIVE	Free Cumulant Sum Gap in Read-Twice BPs for IP_2
2026-05-12 17:56 UTC	INCONCLUSIVE	Tropical Variety Dimension Gap in Read-Twice vs Read-Once BPs
2026-05-12 18:08 UTC	INCONCLUSIVE	Real Rank of SOS Moment Matrices for Max-CUT Bounded by Degree
2026-05-12 18:22 UTC	INCONCLUSIVE	Submodular Rank Gap in Monotone DNF for k-CLIQUE
2026-05-12 19:26 UTC	INCONCLUSIVE	Euler Characteristic of Communication Graph Equals Deterministic
2026-05-12 19:34 UTC	INCONCLUSIVE	Completely Bounded Norm Gap in Read-Twice BPs for IP_2
2026-05-12 19:42 UTC	INCONCLUSIVE	Non-Abelian Fourier Coefficient Spread Inversely Proportional to

Time	Verdict	Title
2026-05-12 20:10 UTC	INCONCLUSIVE	Secant Rank of Disjointness Communication Matrix
2026-05-12 20:40 UTC	INCONCLUSIVE	Lower-Bounds Ran Schur Coefficient Sum Ratio in Symmetric Powers of
2026-05-12 21:10 UTC	INCONCLUSIVE	Permanent vs D Hook-Length Ratio Bounds Monotone Circuit Size for Permanent

How to read the reports

- reports/supported_findings.md — SUPPORTED conjectures with full test code. Paper-quality.
- reports/falsified_findings.md — FALSIFIED with counterexamples.
- reports/daily_YYYY-MM-DD.md — chronological log by day.
- pvsnp_notebook.jsonl — raw JSONL, one entry per cycle (source of truth).

PDF export

Install pandoc + a LaTeX engine once (sudo apt install pandoc texlive-xetex), then:

```
cd research/reports
for f in supported_findings.md falsified_findings.md notebook_summary.md; do
    pandoc "$f" -o "${f%.md}.pdf" --pdf-engine=xelatex -V geometry:margin=2cm
done
```